QUBO Formulation: LLM Cascade Routing

LLM Cascade Routing

Problem Statement. Given a set of user requests $\mathcal{R}$ and a cascade of available language models $\mathcal{M}$, the goal is to assign each request to exactly one model tier. Each assignment (r, m) incurs an inference cost $C_{r,m}$ and yields a predicted quality score Q_m. Each request r has a minimum quality requirement T_r. The objective is to minimize the total inference cost while ensuring that every assigned model satisfies the quality threshold of its request.

Decision Variables. We introduce binary variables $x_{r,m} \in \{0, 1\}$ for all $r \in \mathcal{R}$ and $m \in \mathcal{M}$, where $x_{r,m} = 1$ if and only if request r is assigned to model m, and 0 otherwise.

QUBO Derivation. We construct a Quadratic Unconstrained Binary Optimization (QUBO) formulation by incorporating quality constraints into the objective function via large penalties and enforcing the routing constraint via a squared penalty term.

Let $\tilde{C}_{r,m}$ denote the effective cost for assignment (r, m), defined as:

$$\tilde{C}_{r,m} = C_{r,m} + P \cdot \mathbb{I}(Q_m < T_r),$$

where P is a sufficiently large penalty constant and $\mathbb{I}(\cdot)$ is the indicator function. This ensures that infeasible assignments (those failing the quality threshold) carry a prohibitively high cost.

Step 1: Objective Function. The minimization objective is:

$$\min \sum_{r \in \mathcal{R}} \sum_{m \in \mathcal{M}} \tilde{C}_{r,m} x_{r,m}.$$

Step 2: Constraints. Each request must be routed to exactly one model:

$$\sum_{m \in \mathcal{M}} x_{r,m} = 1, \quad \forall r \in \mathcal{R}.$$

Step 3: Penalty Terms. We enforce the equality constraints using a penalty weight $\lambda > 0$. For each request r, the penalty term is:

$$H_r^{\text{one-hot}} = \lambda \left(\sum_{m \in \mathcal{M}} x_{r,m} - 1 \right)^2.$$

Expanding the square and using the idempotent property $x_{r,m}^2 = x_{r,m}$ for binary variables:

$$H_r^{\text{one-hot}} = \lambda \left(\sum_{m \in \mathcal{M}} x_{r,m}^2 - 2 \sum_{m \in \mathcal{M}} x_{r,m} + 1 \right) \tag{1}$$

$$= \lambda \left(\sum_{m \in \mathcal{M}} x_{r,m} - 2 \sum_{m \in \mathcal{M}} x_{r,m} + 1 \right) \tag{2}$$

$$= \lambda \left(1 - \sum_{m \in \mathcal{M}} x_{r,m} \right). \tag{3}$$

Alternatively, retaining the quadratic form for explicit QUBO structure:

$$H_r^{\text{one-hot}} = \lambda \left(1 - \sum_{m \in \mathcal{M}} x_{r,m} + 2 \sum_{m < m' \in \mathcal{M}} x_{r,m} x_{r,m'} \right). \tag{4}$$

Step 4: Hamiltonian. The total QUBO Hamiltonian $H(x)$ combines the objective and all penalty terms:

$$H(x) = \sum_{r \in \mathcal{R}} \sum_{m \in \mathcal{M}} \tilde{C}_{r,m} x_{r,m} + \sum_{r \in \mathcal{R}} \lambda \left(1 - \sum_{m \in \mathcal{M}} x_{r,m} + 2 \sum_{m < m' \in \mathcal{M}} x_{r,m} x_{r,m'} \right). \tag{5}$$

Step 5: Q Matrix and Offset. Collecting terms, the QUBO parameters are given by:

$$Q_{(r,m),(r,m)} = \tilde{C}_{r,m} - \lambda, \quad \forall r \in \mathcal{R}, m \in \mathcal{M}, \tag{6}$$

$$Q_{(r,m),(r,m')} = 2\lambda, \quad \forall r \in \mathcal{R}, m, m' \in \mathcal{M}, m \neq m', \tag{7}$$

$$Q_{(r,m),(r',m')} = 0, \quad \forall r, r' \in \mathcal{R}, r \neq r', \text{ or } m = m'. \tag{8}$$

The constant offset is:

$$c = \sum_{r \in \mathcal{R}} \lambda = |\mathcal{R}| \lambda.$$

Penalty Weight Justification. The penalty weight λ must be chosen such that any violation of the routing constraint is strictly more costly than the maximum possible reduction in the objective function. Let $W = \max_{r,m} |\tilde{C}_{r,m}|$. If a solution violates a constraint, the penalty incurred is at least λ. The maximum gain in the objective from such a violation is bounded by W. Therefore, setting $\lambda > W$ ensures that the energy of any infeasible solution exceeds that of any feasible solution with non-negative cost, guaranteeing that the global minimum corresponds to a feasible assignment.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^{|\mathcal{R}||\mathcal{M}|}$ recovers the optimal solution because: (i) for any feasible assignment, the penalty terms vanish, and $H(x)$ equals the total effective cost; (ii) for any infeasible assignment, at least one constraint is violated, incurring a penalty of at least λ. Since $\lambda > W$, the energy of any infeasible solution is strictly greater than the maximum possible cost of a feasible solution (assuming non-negative costs). Thus, the global minimum of $H(x)$ must be feasible and minimize the total cost.

Illustrative Example. Consider an instance with $|\mathcal{R}| = 2$ requests and $|\mathcal{M}| = 2$ models. Let the costs be $C_{0,0} = 10$, $C_{0,1} = 20$, $C_{1,0} = 15$, $C_{1,1} = 5$. Let quality scores be $Q_0 = 5$, $Q_1 = 8$, with thresholds $T_0 = 6$, $T_1 = 4$. The assignment $(0, 0)$ is infeasible since $Q_0 < T_0$. Setting $P = 1000$, we have $\tilde{C}_{0,0} = 1010$ and $\tilde{C}_{r,m} = C_{r,m}$ otherwise. Let $\lambda = 1000$.

The effective costs are $\tilde{C}_{0,0} = 1010$, $\tilde{C}_{0,1} = 20$, $\tilde{C}_{1,0} = 15$, $\tilde{C}_{1,1} = 5$. The Q matrix entries are:

$$Q_{(0,0),(0,0)} = 1010 - 1000 = 10, \tag{9}$$
$$Q_{(0,1),(0,1)} = 20 - 1000 = -980, \tag{10}$$
$$Q_{(1,0),(1,0)} = 15 - 1000 = -985, \tag{11}$$
$$Q_{(1,1),(1,1)} = 5 - 1000 = -995, \tag{12}$$
$$Q_{(0,0),(0,1)} = 2000, \quad Q_{(1,0),(1,1)} = 2000, \tag{13}$$

with all other off-diagonal entries zero. The offset is $c = 2 \times 1000 = 2000$.

The Hamiltonian is:

$$H(x) = 10x_{0,0} - 980x_{0,1} - 985x_{1,0} - 995x_{1,1} + 2000x_{0,0}x_{0,1} + 2000x_{1,0}x_{1,1} + 2000. \tag{14}$$

The optimal assignment is $x_{0,1} = 1$, $x_{1,1} = 1$ (requests routed to models 1 and 1), yielding energy $H = -980 - 995 + 2000 = 25$, which matches the total cost $20 + 5 = 25$.